



Sustainable Energy for Financial Inclusion: Optimising Hybrid Energy Systems in Underserved Regions

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Abstract

Access to financial services in rural and underserved regions is hindered by unreliable electricity and limited infrastructure, deepening economic exclusion. This study assesses the potential of a hybrid solar photovoltaic (PV)-diesel generator (DG) system to enhance energy reliability for rural cash centres, using Giedam, a commercial hub in North-East Nigeria, as a case study. HOMER software was used to compare three energy scenarios: diesel-only, PV-battery, and hybrid PV-battery-DG. The hybrid system proved optimal, achieving 85% renewable energy use, reducing diesel consumption by 88%, and maintaining low excess electricity (6.63%), with a competitive cost of energy (\$0.321) and net present cost (\$137,736.10). Solar contributed 87.4% of energy supply, while batteries offered 10.93 h of autonomy. Beyond technical and economic performance, the findings support policy directions that integrate energy planning with financial inclusion strategies. The study underscores the need for targeted subsidies, public-private partnerships, and rural electrification initiatives that prioritise hybrid systems to foster inclusive economic development and environmental sustainability.

Keywords Financial inclusion · Hybrid energy systems · Energy access · HOMER software · Renewable energy · Sustainable development · Techno-economic analysis

Abbreviations

ATM	Automated teller machines
COE	Cost of Energy
DG	Diesel generator
FI	Financial institutions
HES	Hybrid energy system
HOMER	Hybrid optimization model for multiple energy resources
HRES	Hybrid renewable energy systems
NPC	Net Present Cost
PV	Photovoltaic
RES	Renewable energy sources

RF	Renewable fraction
STC	Standard test conditions
P_{PV}	Power generated by the photovoltaic array
f_{PV}	Derating factor of PV
G_T	Solar radiation incident on PV
G_{T-STC}	Incident radiation at STC
α	Temperature coefficient of power
T_c	Temperature in the present timeframe
T_{c-STC}	Temperature of PV cell at STC
P_{max}	Maximum output power
η_{conv}	Converter efficiency
P_{output}	Converter output power
P_{input}	Converter input power
C_{batt}	Capacity of stored energy in the battery
L_D	Load demand or power consumed by the load
B_A	Battery autonomy
$\eta_{batt,rt}$	Round-trip efficiency of the battery system
DoD	Depth of discharge
V_{DC}	Nominal DC voltage of the system
E_{batt}	Energy stored in the battery bank
E_{in}	Energy input to the battery during charging
E_{out}	Energy output from the battery during

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	discharging
η_c	Battery charging efficiency
η_d	Battery discharging efficiency
$SoC(t)$	State of charge of the battery at current time step
$SoC(t-1)$	State of charge of the battery at previous time step
$P(t)_c$	Power input to the battery
$P(t)_d$	Power output from the battery
$C_{batt,min}$	Battery minimum capacity
$C_{batt,max}$	Battery maximum capacity
DoD_{max}	Maximum depth of discharge as defined within
SoC_{min}	Minimum state of charge of battery
SoC_{max}	Maximum state of charge of battery
F	Diesel generator fuel consumption rate
$F_{0,DG}$	Diesel generator fuel curve intercept coefficient
$F_{1,DG}$	Diesel generator fuel curve slope
Y_{DG}	Rated capacity of diesel generator
P_{DG}	Power output from the diesel generator
E_{total}	Total emissions
F_{DG}	Diesel fuel consumed
P_{Grid}	Total grid purchases
EF_{DG}	Emissions factor of diesel generator
EF_{Grid}	Emissions factor of the grid
f_{excess}	Fraction of excess electricity produced
E_{excess}	Excess electricity not consumed or stored
$E_{production}$	Total electricity production (kWh/yr)
E_{served}	Primary electrical load served per year (kWh/year)
$C_{ann,tot}$	Total annualised cost (\$/year)
C_{NPC}	Net present cost
CRF	Capital recovery factor
i	Real discount rate
N	Project lifetime

1 Introduction

Sustainable economic development hinges on ensuring inclusive growth, particularly in the global south countries where equitable access to resources remains a pressing challenge (Lee et al. 2023). Financial inclusion, which involves providing access to essential financial services regardless of location or education level, has emerged as a globally recognised tool for achieving this equitable development (Khan et al. 2023). By addressing both economic and non-economic barriers that perpetuate poverty and inequality, policymakers and stakeholders aim to expand access to crucial services such as risk management, savings, and credit (Asuming et al. 2019; Kebede et al. 2021). In Nigeria's centralised policy

structure, financial institutions (FIs), particularly commercial banks, occupy a central position in economic activity (Jungo et al. 2022). However, these institutions predominantly concentrate their services in urban centres, where they serve a largely literate population through deposit management, loans, and merchant payments (Ozili 2020). This urban focus leaves rural areas significantly underserved, a disparity driven by political neglect, challenging business environments, weak local economies, and limited financial literacy (Ozili 2020; Soetan and Mogaji 2024). As a result, this rural-urban divide continues to impede broader economic participation and sustainable growth across the country.

The challenge of extending financial services to underserved rural areas is compounded by infrastructure limitations, most notably inadequate power supply. Adequate and reliable power supply is fundamental to the operations of FIs, particularly those aiming to serve remote or underserved regions. In many rural areas, where national grid access is either unavailable or highly unreliable, FIs often operate through cash centres, and simplified service outlets that cater to the cash-based economies of these communities. To support the energy needs of such centres, diesel generators (DGs) have long been the dominant solution, serving as either primary or backup power sources (Nsafon et al. 2020; Hosseini Dehshiri and Firoozabadi 2024). These generators have played a critical role in enabling financial services and improving livelihoods in rural settings. However, the continued reliance on DGs poses growing challenges. Rising fuel costs, high maintenance and operational expenses, logistical difficulties in transporting fuel, and environmental concerns related to emissions have rendered DG-based solutions increasingly unsustainable. These pressures, coupled with global efforts to combat climate change, have catalysed a shift toward renewable energy sources (RES) such as solar photovoltaic (PV), wind, micro-hydro, and hybrid energy systems.

In response, many institutions and communities are adopting RES either as direct replacements for DGs, as backup alternatives, or in hybrid configurations that integrate multiple energy sources (Kusakana and Vermaak 2014; Rodríguez-Gallegos et al. 2018; Faraji et al. 2019; Dimitriou and Tsujimura 2019). Optimising these hybrid energy systems (HES) is crucial to achieving cost-effective, reliable, and sustainable energy supply, particularly in rural applications (Kotb et al. 2020). Several studies have investigated such transitions. Ayodele et al. (2019), for example, demonstrated the viability of a hybrid mini-grid combining solar PV, DGs, and battery storage to reduce fossil fuel dependency and emissions in rural Nigerian communities. Similarly, Ukoima et al. (2024) used HOMER software to evaluate four HES configurations for electrifying

Okorobo-Ile, Nigeria, incorporating solar, wind, battery, and fuel cell technologies to determine the most technically and economically feasible option. The importance of tailored energy solutions is further emphasised by (Suresh et al. 2020), which underscore the need for context-specific planning that addresses both financial and environmental goals in remote rural areas. Other global examples reinforce this point. Abid et al. (2021) compared off-grid and grid-connected systems, highlighting the cost benefits of pumped hydro storage in PV-based electrification. In Indonesia, Syahputra and Soesanti (2021) demonstrated the effectiveness of micro-hydro and solar PV systems in bridging electricity supply gaps in underserved regions like rural Yogyakarta Indonesia.

For FIs seeking to expand their rural footprint and advance financial inclusion, these findings underscore the importance of rethinking energy strategies. Transitioning to renewable or hybrid systems offers a path toward resilient, eco-friendly, and economically viable energy solutions. However, before investing in such transitions, it is essential for FIs to evaluate the technical, economic, and environmental implications of various energy configurations. These assessments ensure informed decision-making that aligns with both service delivery goals and sustainability targets, which ultimately strengthen financial access in rural communities.

As one of the most prominent RES, solar photovoltaic (PV) systems have been extensively studied for their technical, economic, and environmental suitability in diverse contexts (Olatomiwa et al. 2015; Sanni et al. 2018). Many of these studies highlight the advantages of hybridising solar PV with complementary sources such as wind energy, using optimal sizing strategies to enhance system performance and sustainability (Olówoséjé et al. 2021; Okundamiya et al. 2022). For instance, Aberilla et al. (2020) assessed a small-scale off-grid PV/wind/battery system designed for rural electrification. The study evaluated 21 configurations, six household systems and 15 microgrids, which found that hybrid PV-wind systems reduced environmental impacts by 17–40% compared to standalone systems. Moreover, integrating battery storage reduced the environmental burden of DGs by 20–30%. The results indicated that combining household-scale PV systems with community-scale wind turbines offers the most sustainable approach for remote communities. Similarly, Ladu et al. (2022) explored the feasibility of a PV/DG/Battery hybrid system in Nyepo Payam, South Sudan, demonstrating improvements in electricity access, cost efficiency, and sustainability. These findings offer valuable guidance for rural energy planning across Africa. In South Africa, however, Mulenga et al. (2023) revealed that extending the national grid to rural areas is prohibitively expensive, positioning distributed microgrids

as a more practical alternative. Although standalone diesel systems may have lower initial capital costs, their high fuel consumption results in the highest long-term energy costs, rendering them economically unviable. To improve reliability in off-grid applications, other studies have integrated advanced storage options such as lithium-ion batteries and flywheels Ngila Mulumba and Farzaneh (2023). Nevertheless, scaling up these solutions requires strong government intervention in the form of enabling policies, targeted investments, and robust regulatory frameworks. Such measures are essential to achieving long-term energy security, environmental sustainability, and equitable energy access in rural areas.

Other studies emphasise the growing role of hybrid renewable energy systems in achieving reliable and sustainable power supply, especially in off-grid and rural settings (Sanni et al. 2019; Amusan et al. 2023; Ennemiri et al. 2024a; Emezirinwune et al. 2024). However, a key design challenge remains the management of excess electricity, which, if unaddressed, leads to inefficiencies and increased costs. Several works report significant surplus generation; for instance, 22% in PV/biogas systems (Ennemiri et al. 2024b) and over 40% in PV/WT/DG/Battery systems (Yakubu et al. 2025). While such excess can support future demand or community growth, its unmanaged existence undermines system efficiency. Roy et al. (2025) demonstrated that more advanced configurations like PV/FC/EL/HT can reduce surplus to below 12%, compared to 54% in simpler PV/BESS setups. Vaziri Rad et al. (2023) however, demonstrated a modified predictive dispatch approach that reduced excess electricity from 41% to 7.5% while utilising 90% of surplus energy and cutting system costs by 10%. In subsequent work, the authors found that larger batteries alone could not reduce excess electricity below 10%, but employing hybrid strategies that combines peak-time fuel usage with hydrogen production achieved under 5% excess electricity.

Building on the existing body of research on hybrid renewable energy systems and their technical optimisation, studies have also explored their application within Nigeria's financial sector. These efforts underscore the potential of HRES to enhance operational efficiency and service delivery in banking institutions. For example, Adesanya and Pearce (2019) presented a compelling business case for private sector investment in hybrid systems, citing clear cost advantages for banking operations. Okundamiya et al. (2014) highlighted the role of reliable energy supply in sustaining banking services such as Automated teller machines (ATM) operations in off-site locations. In a comparative study, Ajide et al. (2021) demonstrated that hybrid PV–diesel systems significantly outperform diesel-only setups in terms of both emissions and cost, though the analysis

focused on urban banks. Similarly, Umoren Mfonobong et al. (2024) evaluated a standalone solar PV system for an ATM gallery, showing how effective load scheduling can support 24-hour operations. While these studies affirm the value of hybrid and renewable energy in banking, they largely concentrate on urban or semi-urban contexts. This leaves a critical gap in the literature regarding the deployment of renewable energy solutions at rural financial service points, particularly cash centres, that are essential to advancing financial inclusion in underserved communities. Table 1 provides a comparative summary of relevant studies and their implications for financial inclusion, underscoring the uniqueness of the present study.

Although HESs integrating DGs have shown clear advantages over diesel-only configurations, such as reduced fuel consumption, extended engine life, and lower operational and environmental costs (Oladigbolu et al. 2020). Prior studies have largely focused on critical public and commercial facilities, including banks, hospitals, and educational institutions (Kolhe et al. 2015; Aziz 2017; Emmanuel et al. 2017; Bashir and Modu 2018; Saiprasad et al. 2018; Kassem et al. 2021; Benti et al. 2022; Ahmed et al. 2023; Ijeoma et al. 2023). However, limited attention has been given to the energy demands of rural financial access points, despite their crucial role in promoting inclusive economic development. In Nigeria, financial inclusion policies have led to a surge in the deployment of digital banking infrastructure, especially ATMs and point of sales machines, but these efforts are heavily skewed toward urban centres and formal

bank branches. Consequently, rural communities remain excluded, contributing to a persistent trust gap and service disparity. To address this gap, the present study introduces a HES model specifically designed for off-grid rural financial service points. The key contributions of this study are summarised as follows:

- Proposing an optimised HES model specifically designed to power rural financial access points, addressing the neglected intersection between energy access and financial inclusion.
- Integrating technical, economic, and inclusion-driven objectives into a unified system design, moving beyond traditional optimisation approaches that overlook social impact.
- Demonstrating the model's application in Geidam, Yobe State, Nigeria, a region characterised by energy poverty and limited financial infrastructure, providing a replicable real-world case study.
- Offering a scalable reference framework for policymakers and investors to align rural electrification initiatives with financial inclusion strategies.
- Bridging a key literature gap by illustrating how decentralised renewable energy solutions can enable last-mile financial services and operational continuity in underserved communities.

Table 1 Comparative summary table

Reference	Energy System Type	Objective	Methods	Application Area	Findings Relevant to Financial Inclusion?
(Adesanya and Pearce 2019)	PV/diesel/storage	Assess viability for different Nigerian industries	HOMER Pro simulations, case studies	Nigerian private industries including banks	None
(Okundamiya et al. 2014)	Hybrid solar/wind/diesel system	Find low-cost setup for powering ATMs	HOMER simulations with local data	ATMs in Ekpoma, Nigeria	Technology-Focused (ATM infrastructure)
(Umoren Mfonobong et al. 2024)	Standalone solar PV	Evaluate PV system for ATM gallery with load scheduling	PVsyst simulations with local data	ATM gallery in Nigeria	Service-Limited (Urban ATM services)
(Ayodele et al. 2019)	Hybrid Solar PV/Battery/Diesel/Wind/Gasoline Generator	Hybrid microgrid optimised to power a microbank and rural homes in remote Nigeria	HOMER-based economic and sensitivity analysis	Ajasse-Ipo village, Kwara State, Nigeria (rural micro-bank+community loads)	Community-Oriented (Rural financial services)
This Study	PV/DG/Battery	Electrify rural cash centre to support financial inclusion	HOMER-based techno-economic optimisation	Rural ATM/cash centre (Geidam)	Comprehensive Impact (Infrastructure, services, community needs, and policy in rural settings)

Table 2 Electrical equipment and power rating within the cash centre used in this study

Equipment	Numbers	Rating (W)	Total (W)	Operation Hours/Day	Daily Energy (Wh)
ATM	6	650	3900	20	78,000
Air conditioner (inverter driven)	7	835	5845	4	17,535
Lightning point (LED)	15	20	300	8	2,400
Lightning point (Others)	9	15	135	6	810
Router	1	70	70	24	1,680
CCTV	10	15	150	24	3,600
Laptop	12	70	840	8	3,360
Network switch	1	150	150	24	3,600
Money counting machines	6	75	450	5	900
Signage	1	600	600	6	3,600
Security system	1	1500	1500	10	18,000
Refrigerators	3	400	1200	8	12,000
Television	1	50	50	4	250
Total			15,190		

2 Materials and methods

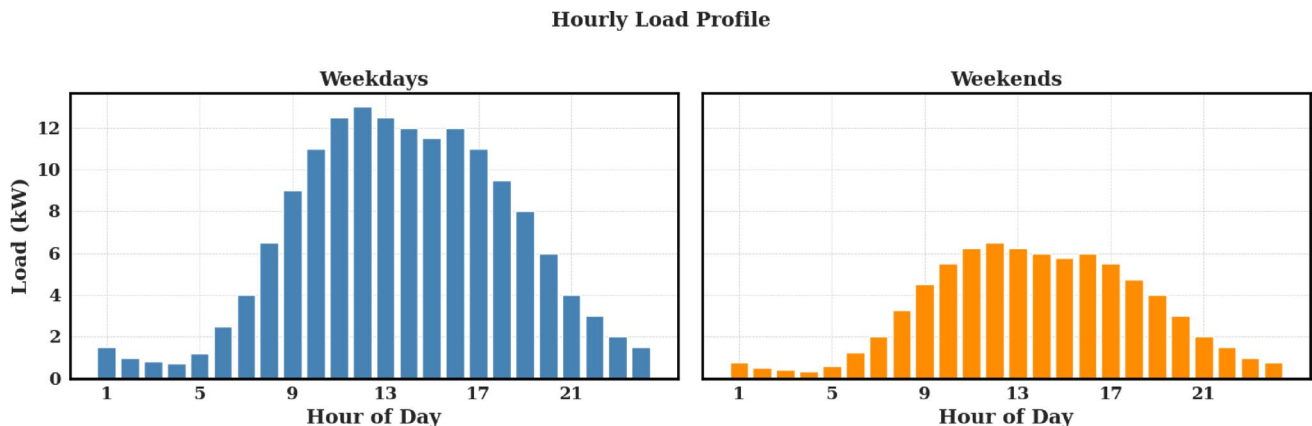
A hybrid solar PV–diesel system is evaluated for its potential to complement the delivery of financial services to a typical rural town in Nigeria. HOMER software simulations were conducted using predefined component specifications to identify the optimal solution.

2.1 Site Description and Load Data

Giedam is a remote commercial town located at latitude 12° 53' 42.13"N and longitude 11° 55' 42.43"E, in Yobe State,

northeastern Nigeria. It is known for its vibrant economic activities, including agriculture, livestock trading, artisanal fishing, and cross-border commerce with neighbouring countries such as Niger, Chad, and Cameroon. These trade linkages position Giedam as one of the most active rural commercial hubs in the North-East region of Nigeria. Population projections from the 2006 national census using a 3% annual growth rate, indicates an estimate of 230,993 people (UNICEF 2024). Despite its demographic and economic significance, Giedam suffers from severe infrastructure deficits—most notably, limited access to reliable electricity. This energy shortfall hampers commercial productivity and essential service delivery. National data indicates that the highest percentage of Nigerians without electricity live in rural areas, particularly in the northern regions (Statista 2024). Giedam exemplifies this trend, with many households lacking access to grid electricity, further deepening the development gap in the region.

As a result of this unreliable grid access, cash-based transactions dominate Giedam's financial landscape, supported by cash centres of commercial banks that serve both residents and cross-border customers. These centres are critical financial points that require uninterrupted electricity to operate security systems, lighting, computers, ATMs, and cooling equipment. Table 2 presents the rated power of the typical equipment used, from which an estimated load profile (Fig. 1) for a standard cash centre was derived. It shows the hourly load distribution for weekdays and weekends at the bank. On weekdays, demand rises around 7:00 a.m., peaks at 12.5 kW by 11:00 a.m., and remains high until 4:00 p.m., reflecting full business operations. Evening demand accounts for residual activities like audits and backups. Weekends are modelled at 50% of weekday values to reflect reduced hours and limited services, consistent with rural banking practices in Nigeria. The average daily consumption is 150 kWh, with a peak connected load of 15.19 kW. However, actual demand is lower due to load

**Fig. 1** Hourly load data of the selected cash centre

diversity. A 5% day-to-day variability was added to reflect operational fluctuations in customer volume and equipment use. This assumption, common in similar commercial models, enhances practicality of the simulation.

2.2 Solar Energy Potential and Resource Mapping

The study area is characterised by a tropical Sahel climate, providing optimal conditions for high solar radiation, making it well-suited for harnessing solar energy. Leveraging this abundant and sustainable resource offers a critical alternative energy supply for the town's cash centres. However, accurately assessing the energy production potential is crucial for evaluating the performance of the proposed system. In this study, data from NASA's solar resource measurements were employed to estimate the energy output of the proposed solar energy system (NASA 2024). Figure 2 illustrates the monthly variation in daily average solar irradiation and the clearness index. The data reveal an annual average daily solar radiation of 6.11 kWh/m², with values ranging from a minimum of 5.27 kWh/m² in December to a maximum of 6.98 kWh/m² in April. These figures underscore the area's strong potential for solar photovoltaic (PV) applications to meet its energy demands.

Additionally, the temperature profile of this semi-arid region, depicted in Fig. 3, shows monthly variations from a low of 21.89 °C in January to a peak of 33.37 °C in May,

with an annual average temperature of 27.89 °C. This temperature range significantly impacts the efficiency and performance of solar PV systems, emphasising the need for careful design and optimisation to ensure reliable energy production.

2.3 Hybrid System Specifications and Mathematical Modelling of the Components

The schematic diagram of the HES considered in this study is presented in Fig. 4. The system comprises key components, including a DG, solar photovoltaic (PV) panels, batteries, and a converter system. This study evaluates the techno-economic implications of the optimal combination of these components for the cash centre, using the Hybrid Optimization Model for Multiple Energy Resources (HOMER) software (version 3.14.2) for system analysis and optimisation. HOMER was chosen due to its extensive use and proven capability to simulate hybrid systems, perform sensitivity and economic analysis, and evaluate a wide range of configurations efficiently (Sinha and Chandel 2014). It also offers user flexibility and detailed modelling of diverse energy sources, making it particularly suitable for off-grid system studies. The modelling of each system component is detailed in the following sections, outlining their technical and operational characteristics. Table 3 summarizes the technical and cost implications of the various components,

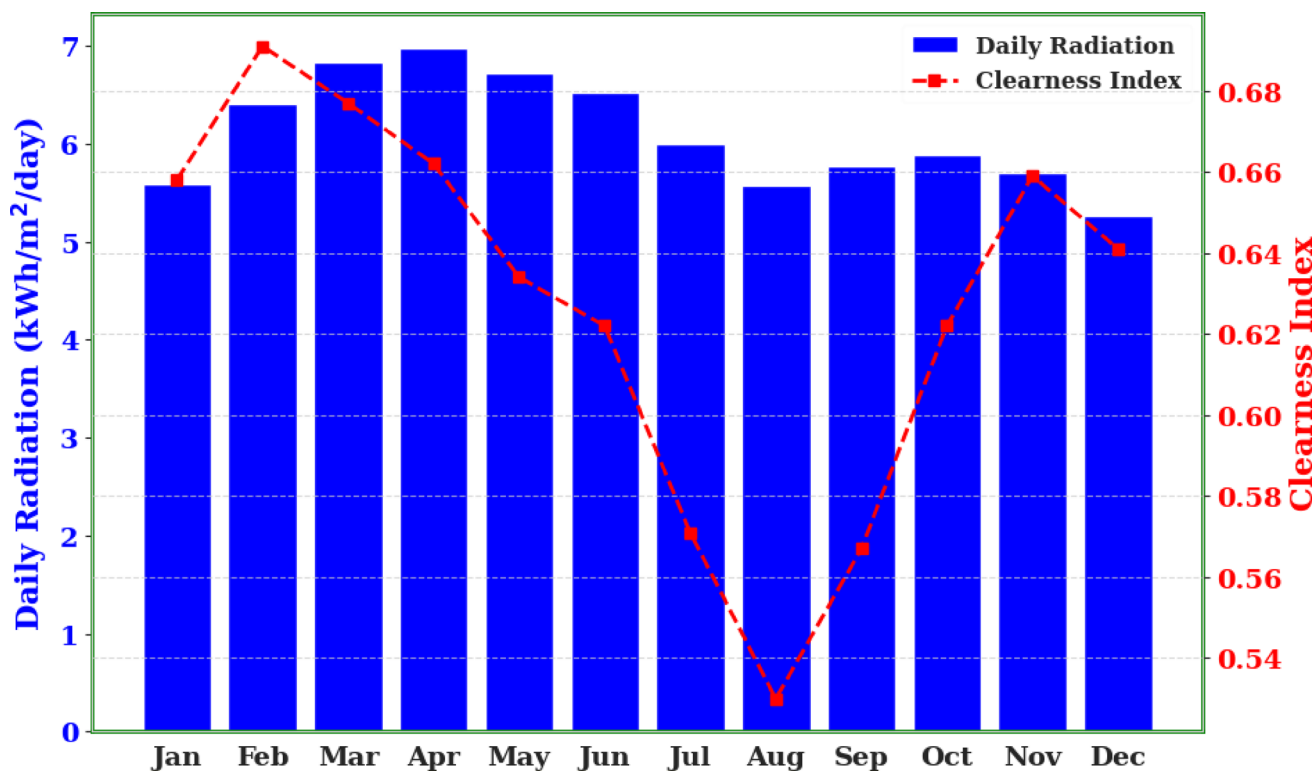


Fig. 2 Monthly solar irradiation and clearness index of study area

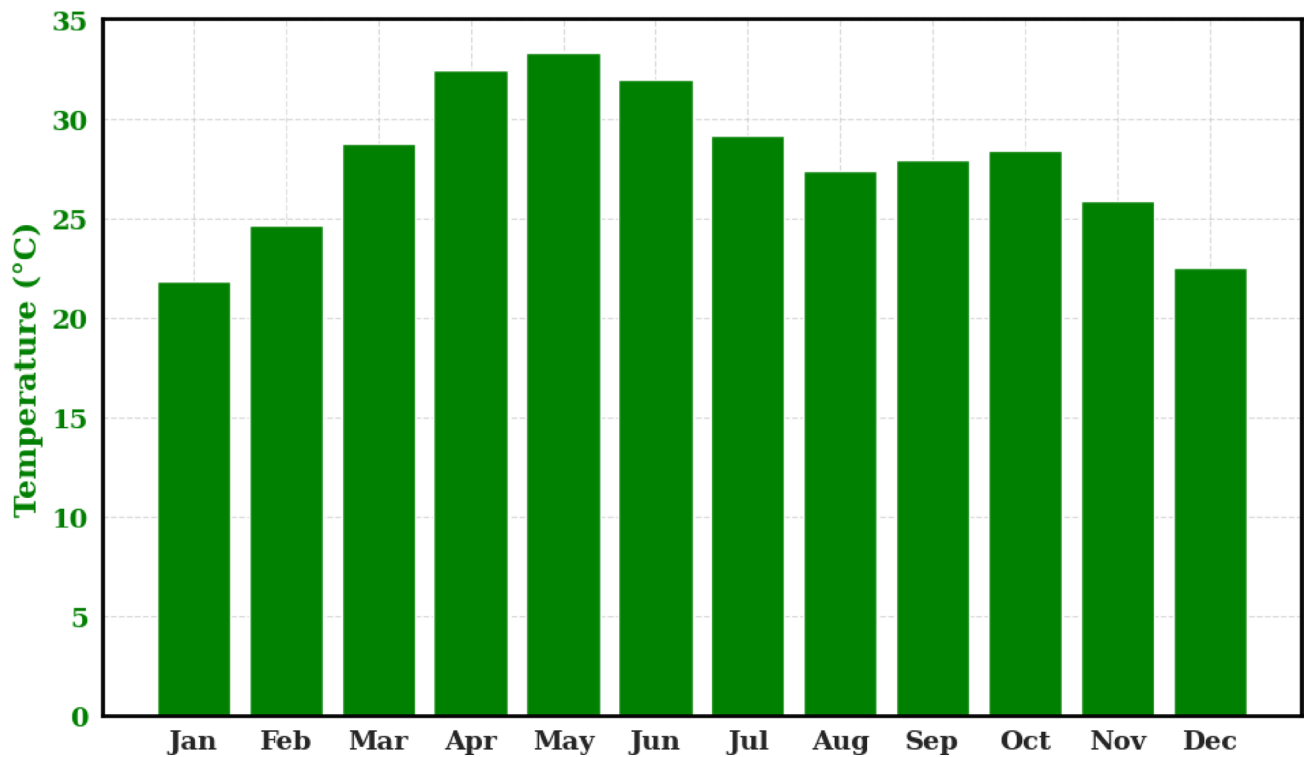
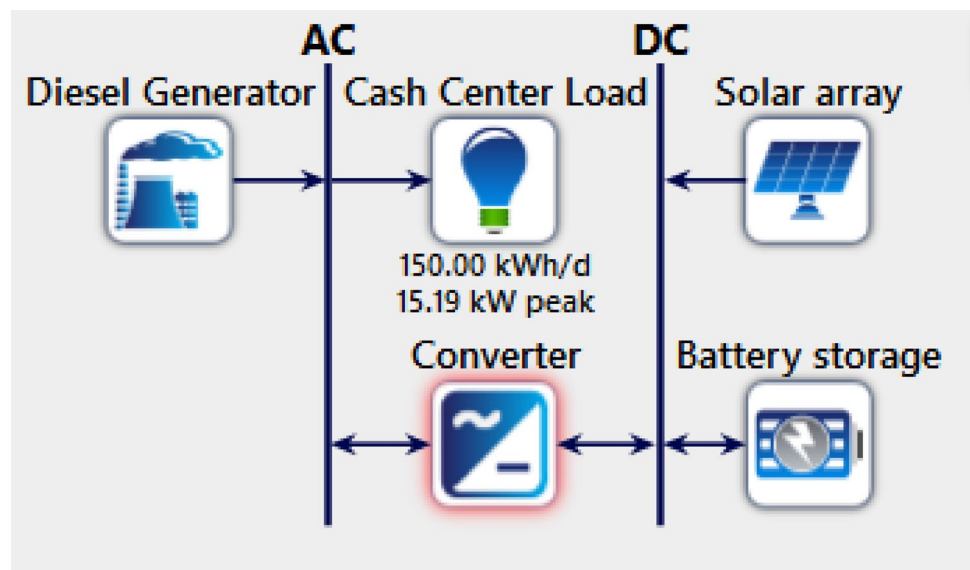


Fig. 3 Monthly mean temperature in study area

Fig. 4 Schematic of the system under study



serving as input parameters for the HOMER analysis. This comprehensive approach ensures a well-informed assessment of the system's performance and economic viability.

2.3.1 Solar PV System

The electrical power generated by a photovoltaic (PV) module is influenced by several factors, including geographical

location, solar resource availability, and the module's operating temperature. For this study, a JINKO solar PV panel with a power rating of 300 W and an efficiency of 18.33% was selected. The cost associated with the panel is detailed in Table 3, and its operating temperature is specified as 45 °C. The power output of the PV module is calculated using the expression in (1) (HOMER Pro):

Table 3 Technical and cost specifications of the required components for system modelling

Component	Parameter	Value	Lifetime	Capital Cost	O&M Cost
PV System (Neeg Power Solutions Limited 2024)	Rated Capacity	0.3 kW	25 years	\$50	\$50/year
	Efficiency	18.33%	—	—	—
	Replacement Cost	\$50	—	—	—
Diesel Generator (EnergyMall 2019)	Rated Capacity	16 kW	15,000 h	\$5,000	\$0.6/Op. Hour
	Fuel Type	Diesel	—	—	—
	Fuel Cost	\$0.93/litre	—	—	—
	Replacement Cost	\$5,000	—	—	—
Battery (Gennex Technologies 2019)	Nominal Voltage	12 V	5 years	\$310	\$3.00/year
	Nominal Capacity	2.63 kWh	—	—	—
	Replacement Cost	\$310	—	—	—
	Throughput	1,845kWh			
Converter (Jumia Nigeria 2024; Mega Stores Limited 2024)	Capacity	1 kW	15 years	\$180	\$0
	Efficiency	95%	—	—	—
	Replacement Cost	\$180	—	—	—

$$P_{PV} = P_{PV} \times f_{PV} \left(\frac{G_T}{G_{T-STC}} \right) (1 - \alpha (T_c - T_{c-STC})) \quad (6)$$

Where, P_{PV} is the rated capacity of the PV, f_{PV} is the derating factor of PV, and G_T is solar radiation incident on PV in the present timeframe. G_{T-STC} represents the incident radiation at STC, α is the temperature coefficient of power considered as 0.5, T_c signifies PV cell temperature in the present timeframe and T_{c-STC} is the temperature of PV cell at STC, considered as 25 °C.

2.3.2 Converter

The converter system facilitates bidirectional energy flow between the DC components and the AC system, ensuring seamless operation of the HES. For this study, both the rectifier and inverter systems are assumed to operate with an efficiency of 95%, and their projected lifetime is 15 years. The associated costs of the power converter are detailed in Table 3. The relationship between the maximum output power (P_{max}) and the conversion efficiency (η_{conv}) is given by the following expression (HOMER Pro):

$$P_{conv} = \frac{P_{max}}{\eta_{conv}} \quad (7)$$

$$\eta_{conv} = \frac{P_{output}}{P_{input}} \quad (8)$$

2.3.3 Battery Storage

The battery storage system in the HES is a crucial component, designed to store energy generated by the PV system for use during periods when other energy sources are limited or unavailable. For this study, a Trojan battery system is employed, with cost details provided in Table 3. To ensure battery health and longevity, a minimum state of charge of 35% is specified, targeting a minimum operational lifespan of 5 years. The capacity of stored energy in the battery (C_{batt}) is calculated using the following equation (HOMER Pro):

$$C_{batt} = \frac{L_D \times B_A}{\eta_{batt,rt} \times DoD \times V_{DC}} \quad (9)$$

Where L_D is the electrical load to be supplied, B_A are the days of battery autonomy, $\eta_{batt,rt}$ is the round-trip efficiency of the battery, DoD is the depth of discharge, and V_{DC} is the nominal voltage of the system (HOMER Pro).

Equation (10) describes the energy added to the battery during charging and the energy available to the load during discharging, both of which are governed by efficiency factors that account for losses in the energy conversion process (HOMER Pro).

$$\left. \begin{aligned} E_{batt} &= \eta_c \times E_{in} \\ E_{out} &= \eta_d \times E_{batt} \end{aligned} \right\} \quad (10)$$

Where η_c and η_d are the battery charging and discharging efficiencies, respectively, while E_{batt} , E_{in} , and E_{out} represent the energy stored in the battery, the energy input to the battery, and the energy drawn from the battery storage, respectively.

In many cases, HOMER simplifies battery performance modelling by using a round-trip efficiency ($\eta_{batt,rt}$), calculated as the product of the battery's charge efficiency and discharge efficiency (HOMER Pro):

$$\eta_{batt,rt} = \eta_c \times \eta_d \quad (11)$$

For each time step of the simulation, HOMER updates the battery's state of charge using Eq. (12):

$$SoC(t) = SoC(t-1) + \frac{P(t)_c \times \eta_c - \frac{P(t)_d}{\eta_d}}{C_{batt}} \quad (12)$$

Where $SoC(t)$ indicates the state of charge at time t ; $P(t)_c$ represents the power input to the battery; $P(t)_d$ is the power output from the battery.

The battery management system's ensure the battery capacity is within the minimum and maximum set limit defined by Eqs. (13) and (14) while the charge/discharge cycle modelled by Eqs. (15) and (16) as (HOMER Pro):

$$C_{batt,min} \leq C_{batt} \leq C_{batt,max} \quad (13)$$

$$C_{batt,min} = (1 - DoD_{max}) \times C_{batt} \quad (14)$$

$$C_{batt,min} = SoC_{min} \times C_{batt} \quad (15)$$

Where $C_{batt,min}$ and $C_{batt,max}$ represents the battery's minimum and maximum capacity, DoD_{max} is the maximum depth of discharge as defined within HOMER, and SoC_{min} is the minimum state of charge at a given time.

As part of measure to prolong the life time of the storage systems, the state of charge of the battery is maintained using (16) (HOMER Pro):

$$SoC_{min} \leq SoC(t) \leq SoC_{max} \quad (16)$$

2.3.4 Diesel Generator

The diesel generator (DG) currently serves as the primary energy source for most commercial operations in the town. However, this study explores integrating the DG into a HES, where it functions as a backup to the solar PV system. A 20kVA DG with low emission in compliance with EU Stage IIIA emissions standards is selected in HOMER (HOMER Pro; EnergyMall 2019). This is aimed at further mitigating environmental pollution caused by emissions

while addressing the challenges posed by the fluctuating and rising cost of diesel fuel. By adopting this hybrid strategy, the cash centre can achieve a more sustainable and long-term energy solution for its operations. The cost components associated with the DG system are detailed in Table 3. The fuel consumption rate (F) of the DG is calculated using Eq. (17) (HOMER Pro):

$$F = F_{0,DG} Y_{DG} + F_{1,DG} P_{DG} \quad (17)$$

where, is the $F_{0,DG}$, $F_{1,DG}$, Y_{DG} , and P_{DG} represents diesel generator's fuel curve intercept coefficient, fuel curve slope, rated capacity, and electrical output of the diesel generator respectively. In HOMER, emissions are calculated using Eq. (18), which combines fuel-derived emissions (based on annual consumption) with indirect grid-purchase emissions. For this study the later component is unaccounted for considering the absence of grid connection (Sanni et al. 2021):

$$E_{total} = EF_{DG} F_{DG} + EF_{Grid} P_{Grid} \quad (18)$$

where, is the E_{total} , F_{DG} , and P_{Grid} represent the total emissions, diesel fuel consumed and total grid purchases. EF_{DG} and EF_{Grid} are the emissions factors of the DG and grid respectively.

2.4 Energy Production Dispatch Strategy

The energy production of the proposed HES requires an optimised dispatch strategy to effectively manage available resources and minimise operational costs. In this study, the predefined combined dispatch strategy in HOMER was selected for its economic efficiency in meeting the system's energy demands. This approach integrates both load-following and cycle-charging methods, dynamically adjusting the operation of system components based on prevailing economic and energy conditions.

Under this strategy, the DG is activated only when the solar PV system and battery storage are unable to meet the load requirements. In cycle-charging mode, the DG operates at full capacity, with any excess energy generated directed to charge the battery system. Conversely, during periods of sufficient solar energy availability, the DG runs only at the level necessary to meet the load demand, prioritising solar energy for charging the batteries. Figure 5 illustrates this dispatch strategy for the selection of the appropriate dispatch strategy the demand to be met.

This dual-mode operation reduces energy wastage, optimises fuel consumption, and lowers fuel costs. Additionally, the strategy supports fuel savings and extends the operational life of the battery system, thereby ensuring a cost-effective and sustainable energy supply for the cash centre.

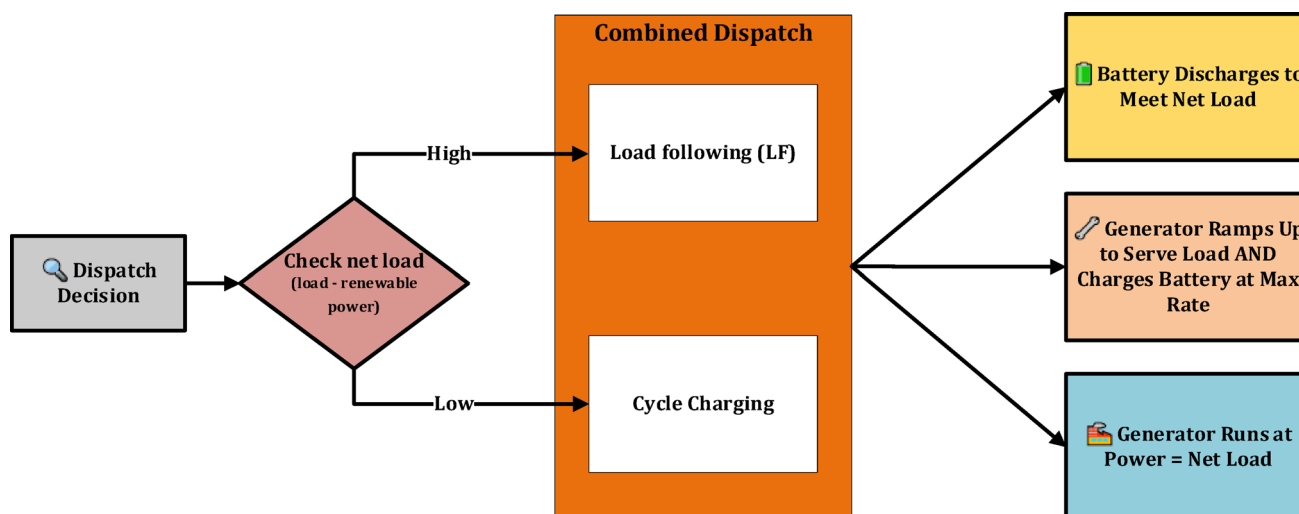


Fig. 5 Workflow diagram of the dispatch strategy based on net load and power component utilisation

HOMER Pro quantifies system performance by calculating the fraction of excess electricity produced at each simulation's conclusion using Eq. (19) (HOMER Pro):

$$f_{\text{excess}} = \frac{E_{\text{excess}}}{E_{\text{production}}} \quad (19)$$

Where E_{excess} gives the total excess electricity (kWh/yr) and $E_{\text{production}}$ represents the total electrical production (kWh/yr).

2.5 Econometric Model

HES projects are financially intensive, often constrained by limited resources. Therefore, understanding the financial implications of such projects is essential to justify their implementation. Econometric models, such as the Cost of Energy (COE), assess the cost-effectiveness of proposed solutions, where a lower COE indicates a more economically viable project. The COE is defined as the cost per unit of energy consumed by the served load and is expressed in Eq. (20) as follows (HOMER Pro):

$$COE = \frac{C_{\text{ann,tot}}}{E_{\text{served}}} \quad (20)$$

where E_{served} is the primary electrical load served per year (kWh/year), and $C_{\text{ann,tot}}$ represents the total annualised cost (\$/year).

In this study, HOMER simulation, ranks optimal combinations of system components based on the Net Present Cost (NPC), which reflects the life cycle cost of the system. This approach considers the cost of all components without accounting for the present value of revenue over the

project's lifetime. The NPC is calculated using Eq. (21) as (HOMER Pro):

$$C_{NPC} = \frac{C_{\text{ann,tot}}}{CRF(i, N)} \quad (21)$$

Where CRF , i , and N represents the capital recovery factor, real discount rate, and project lifetime respectively. The CRF , shown in Eq. (22), calculates the present value of an annuity and is defined as (HOMER Pro):

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (22)$$

2.6 Sensitivity Parameters

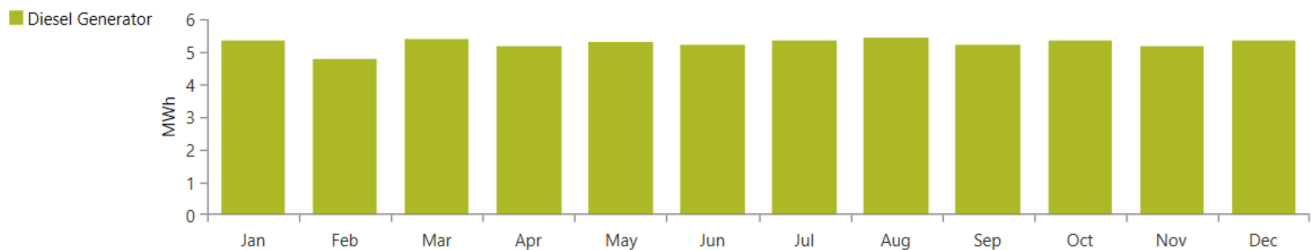
Certain system parameters may change over the course of the project's lifetime, impacting optimal configuration performance. Therefore, it is essential to investigate the effects of these changing variables on system performance. This study examines the sensitivity of the most optimal configuration to variables such as diesel prices and solar radiation levels. These variations include both increases and decreases around the base values, providing an overview of how such changes could affect the system's overall techno-economic performance.

3 Results and Discussion

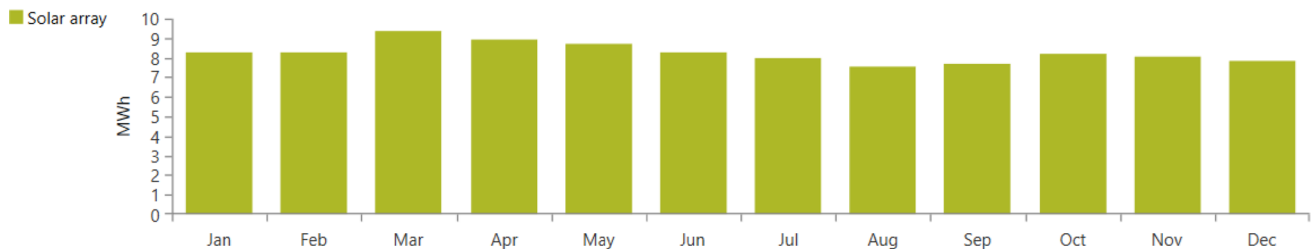
Using HOMER simulation, the study evaluated the load profile and energy resources of Geidam town, Nigeria, to determine the most efficient HES configuration. The

Table 4 Optimal feasible solutions and components sizing

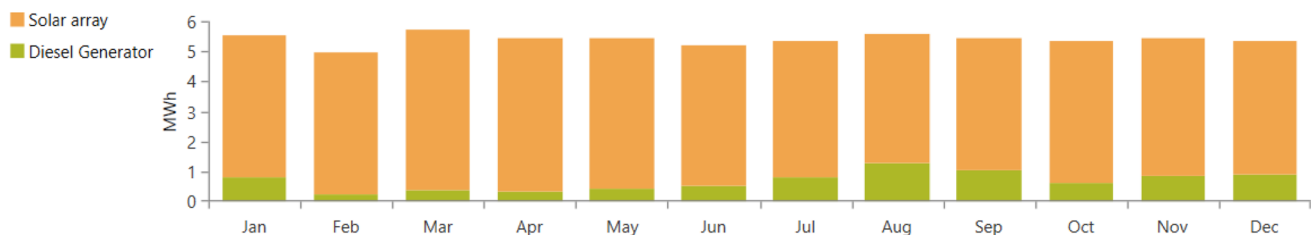
Scenario	Configuration	Diesel			PV capacity (kW)	Battery (Nos)	Converter (kW)	Renewable fraction (%)
		Capacity (kW)	Total Fuel (L/yr)	Operating hours				
I	DG	16	22,719	8760	-	-	-	0
II	PV/Battery	-	0	-	78	40	16	100
III	PV/Battery/DG	16	2740	659	48	40	16	85



(a): Scenario I



(b): Scenario II



(c): Scenario III

Fig. 6 Monthly electricity production for each scenario

assessment included a comparative analysis of multiple configurations, factoring in techno-economic feasibility, environmental impacts and sensitivity analysis. HOMER executes an hourly resolution simulation to determine the most feasible configuration using already defined input parameters. Based on the solar resource's potential and load requirement of the cash centre, three feasible optimal configurations were thrown up as being feasible (Table 4). For each of this configuration, analysis is presented based on the technical, economic and environmental implications. In addition, the sensitivity of these results to different variables is also evaluated.

3.1 Technical Assessment

The comparative analysis of the three optimal configurations reveals their strengths, limitations, and relative performance across technical and operational metrics. Figure 6 illustrates the monthly electricity production for all scenarios. It is apparent that in scenario I (DG-only), electricity production remains mostly constant throughout the year, with the exception of a noticeable variation in February. Scenario II indicate total electricity production from the solar PV system alone, with the lowest output observed in August and the highest in March. In Scenario III (PV/Battery/DG), the DG contributed least in the month of February in contrast to

the month of August where its contribution was higher due to lower solar production.

Furthermore, Table 5 presents the performance of the three HES configurations based on capacity shortage, electricity production, excess electricity, unmet load, and individual component contributions. The DG-only configuration (Scenario I) generated a total of 63,079.95 kWh/year, consistently meeting the energy demands of the facility with no capacity shortage or unmet load throughout the year. This demonstrates the reliability of DGs in providing stable and continuous power, particularly in off-grid settings. However, the system recorded a relatively high 13.21% excess electricity, indicating inherent inefficiencies typical of standalone diesel systems. These inefficiencies stem from the generator's fixed output characteristics and its operation at partial loads during low-demand periods, since fuel consumption does not necessarily scale proportionally with energy output. The complete reliance on fossil fuel generation implies higher emissions, greater exposure to fuel price volatility particularly for remote areas, and increased maintenance demands, making this configuration less sustainable and more costly over time compared to hybrid or renewable alternatives.

In contrast, the PV/Battery system (Scenario II) produced a total of 91,934.56 kWh/year, representing a 46% increase in energy output compared to the DG-only configuration. Despite this apparent advantage in renewable generation, the system experienced a 2.85% capacity shortage and a significantly high 36% excess electricity, suggesting inadequate energy utilisation. This excessive surplus is primarily attributed to the PV array's sizing strategy, which was designed to independently meet all load demands throughout the year. However, due to inadequate battery storage capacity arising from the need to optimise system cost, a substantial portion of the solar energy could not be stored or used when demand was low or generation was high, particularly during midday hours. As a result, a significant amount of generated electricity was unused. Even at that, the system's battery bank provided 10.93 h of autonomy, which reflects a relatively strong storage capacity for short-term backup. Nevertheless, it still recorded 2.14% unmet load, highlighting reliability limitations during extended periods of low solar irradiance or when back-to-back cloudy days reduced PV output. These periods of mismatch between energy availability and demand, coupled with no backup from a generator, rendered

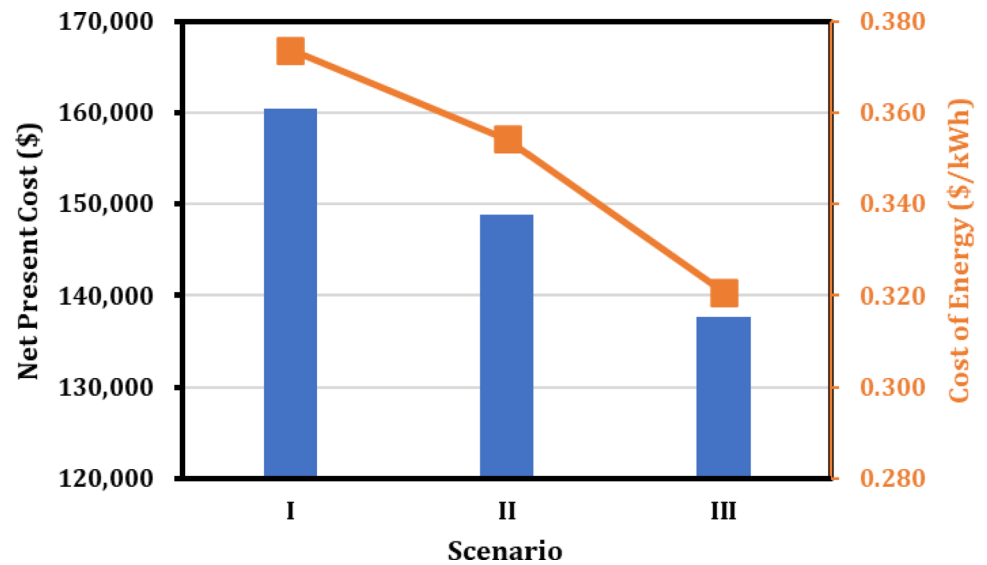
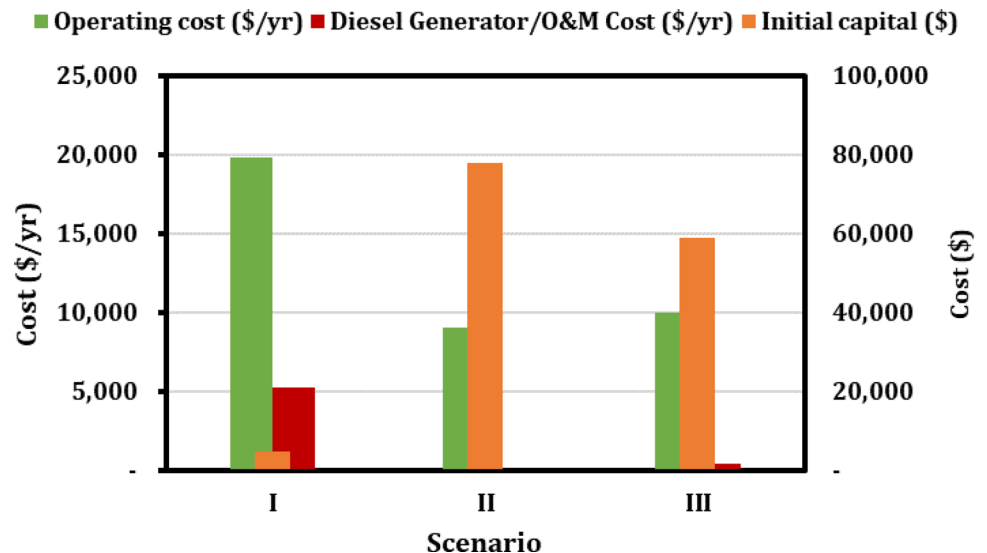
the system vulnerable to shortfalls, especially during critical load periods. This highlights the need for enhanced energy management strategy and a more balanced configuration in standalone renewable systems, where energy generation, storage, and consumption must be optimally aligned to ensure both efficiency and reliability.

The hybrid PV/DG/Battery configuration (Scenario III) exhibited the most balanced and resilient performance among all evaluated scenarios. It generated a total of 64,756.80 kWh/year, effectively satisfying the system's annual energy demand with zero capacity shortage and no unmet load throughout the simulation period. This reliability can be attributed to the strategic integration of solar PV, battery storage, and diesel backup, which allowed the system to make the most of renewable energy availability while compensating for its intermittency. In this configuration, solar PV contributed 56,575.12 kWh/year, accounting for 87.4% of the total electricity production. The DG produced 8,181.68 kWh/year, playing a crucial role in bridging shortfalls during periods of low solar irradiance or high load demand, particularly at night or during overcast conditions. By operating the DG-only, when necessary, the system significantly reduced fuel consumption and operational costs, while still ensuring reliability. Importantly, excess electricity was minimised to 6.63%, a significant improvement over the 36% observed in the PV/Battery-only system. This indicates a well-coordinated energy flow, where generation more closely matches demand, and surplus energy is largely avoided through intelligent dispatch strategies. Although net metering could theoretically absorb excess solar generation during low-demand periods (e.g., weekends or holidays), this option is constrained in the study area due to widespread grid unreliability. In such contexts, surplus energy that cannot be exported is redirected to other non-critical or deferrable loads (such as water pumping) where available to avoid energy wastage. The system also maintained a battery autonomy of 10.93 h, identical to Scenario II, but with markedly better reliability and efficient energy utilisation. Overall, Scenario III demonstrates a superior synergy between components, achieving an optimal trade-off between sustainability, reliability, and efficiency. It highlights the value of hybridisation in off-grid applications, particularly in critical sectors like FIs, where uninterrupted power supply is essential, but long-term operational costs and fuel dependence must also be minimised.

Table 5 Energy production outputs of each scenario

Scenario	Configuration	Cap short (%)	Elec Prod (kWh/yr)	Excess Elec (%)	Unmet load (%)	Production (kWh/yr)		Battery Autonomy (hr)
						Diesel generator	Solar array	
I	DG	0	63,079.95	13.21	0	63,079.95	0	0
II	PV/Battery	2.85	91,934.56	36	2.14	0	91,934.56	10.93
III	PV/Battery/DG	0	64,756.80	6.63	0	8,181.68	56,575.12	10.93



Fig. 7 NPC and COE for each scenario**Fig. 8** Initial capital and operating cost for each scenario

3.2 Economic Analysis

Evaluating the economic attributes of each configuration showcases their alignment with key indices like NPC and COE. NPC integrates initial costs, replacement expenditures, O&M, fuel costs, and salvage value. The financial performance of the three configurations reveals significant contrasts as shown in Figs. 7 and 8. The hybrid PV/Battery/DG system demonstrated the best economic outcomes with the lowest NPC of \$137,736.10 and the lowest COE at \$0.321. Its operating cost was moderate at \$10,032.42/year, while the initial capital investment was \$59,059.90. This balance between upfront costs and operational efficiency highlights the long-term cost benefits of hybrid systems. In Scenario II, the PV/Battery configuration ranked second, with an NPC of \$148,859.10 and a COE of \$0.354. It had the lowest operating cost of \$9,027.97/year due to

the absence of diesel consumption. However, its initial capital requirement of \$78,059.90 was the highest, reflecting the significant investment needed for renewable energy systems and battery storage. The DG-only configuration (Scenario I) had the highest NPC at \$160,429.80 and the highest COE of \$0.374, driven by its substantial operating costs of \$19,819.67/year; 71% of which was spent on diesel fuel. While it required the lowest initial capital of \$5,000, its dependency on diesel fuel resulted in high recurring expenses, making it the least cost-effective option over time.

The comparative nominal cash flow for the three scenarios is illustrated in Fig. 9. Scenario I (DG-only) demonstrates a consistent and relatively higher cash outflow over the entire project lifetime, primarily driven by ongoing fuel consumption and maintenance costs associated with continuous DG operation. This steady financial burden makes the DG-only configuration less attractive in the long run,

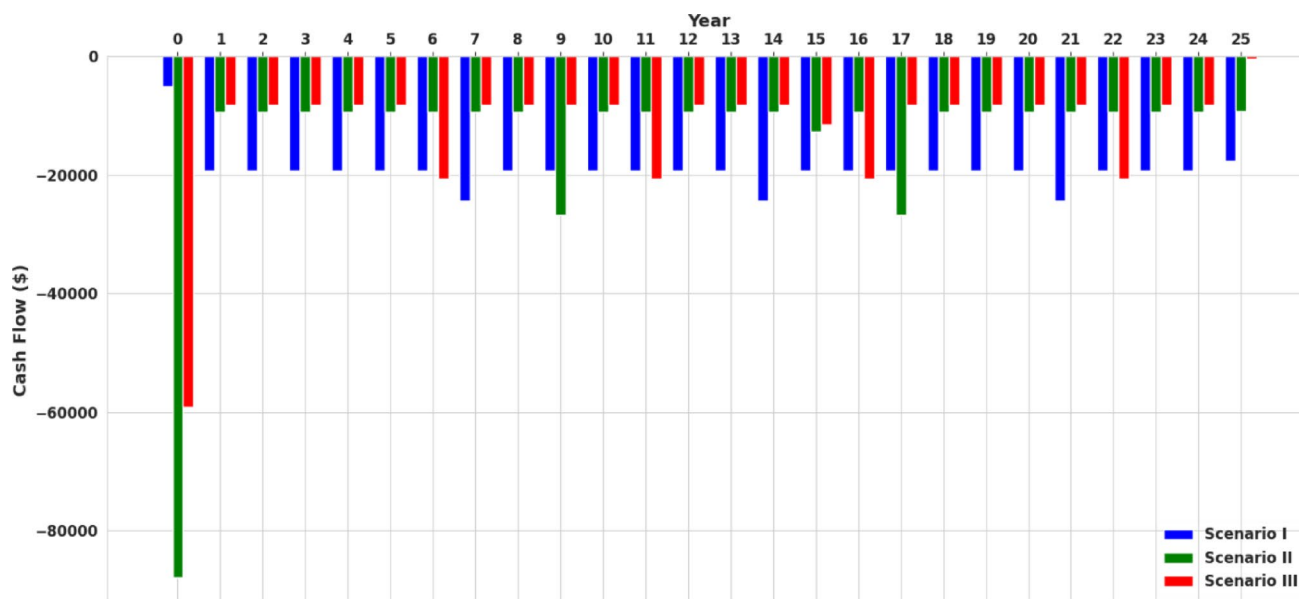


Fig. 9 Comparative nominal cash flow for each scenario

particularly under fluctuating fuel prices and increasing environmental concerns.

In contrast, Scenario II (PV/battery) and Scenario III (PV/battery/DG) exhibit lower cash flow in the initial five years, reflecting reduced dependency on fossil fuels and lower operational costs due to the incorporation of renewable energy and storage technologies. However, both hybrid configurations show intermittent spikes in cash flow, most notably every 6 to 10 years, which correspond to scheduled battery replacements. These spikes highlight the impact of battery degradation over time, which is a critical factor in hybrid system planning. Battery lifespan is influenced by factors such as depth of discharge, charge/discharge cycles, ambient temperature, and overall system usage. As batteries degrade, their effective storage capacity diminishes, leading to replacement needs that significantly influence the NPC of the system.

Despite this, Scenario III still retains a clear economic edge over both the DG-only and PV/battery configurations across the project timeline. This advantage is due to its operational flexibility and ability to optimise dispatch between the generator and battery. By leveraging the generator only when solar and battery sources are insufficient, Scenario III reduces excessive cycling and degradation of the battery, thereby extending its lifespan and deferring replacement costs. Furthermore, the inclusion of the DG in Scenario III helps buffer against solar intermittency and reduces the reliance on deep battery discharges, two major contributors to accelerated battery wear. This hybrid approach not only offers a technical advantage but also supports long-term cost savings through balanced asset utilisation. Though battery degradation challenges HESs, its impact is manageable

through appropriate system design, scheduled maintenance, and optimised dispatch strategies. Scenario III exemplifies a balanced solution that addresses both technical and economic concerns, reinforcing its suitability for rural off-grid financial service applications.

This techno-economic performance must also be understood within the broader energy access context. Although Nigeria's nominal grid tariff of \$0.04/kWh assumes 12 h of daily electricity supply, this expectation remains unmet in Geidam where power is largely unavailable. Scenario III's higher COE of \$0.321/kWh provides a realistic cost benchmark for reliable hybrid power delivery, incorporating all capital, replacement, and maintenance expenses. In a region dominated by diesel generators due to grid unreliability, this figure represents the most economically justified solution compared to diesel-only systems or dependence on an unreliable grid.

3.3 Environmental Implications

The environmental impacts of the different energy system configurations are evaluated to emphasise the role of renewable energy systems in reducing pollution and supporting sustainable development. As shown in Fig. 10, the DG-only configuration (Scenario I) presented the highest environmental burden, emitting 60,114.05 kg of CO₂ annually due to continuous diesel generator operation. In contrast, the hybrid PV/Battery/DG system (Scenario III) significantly reduced emissions to 7,249.87 kg of CO₂ per year. This reduction was achieved through strategic minimisation of diesel usage, with the DG accounting for only 15% of the system's total energy production. In addition to carbon

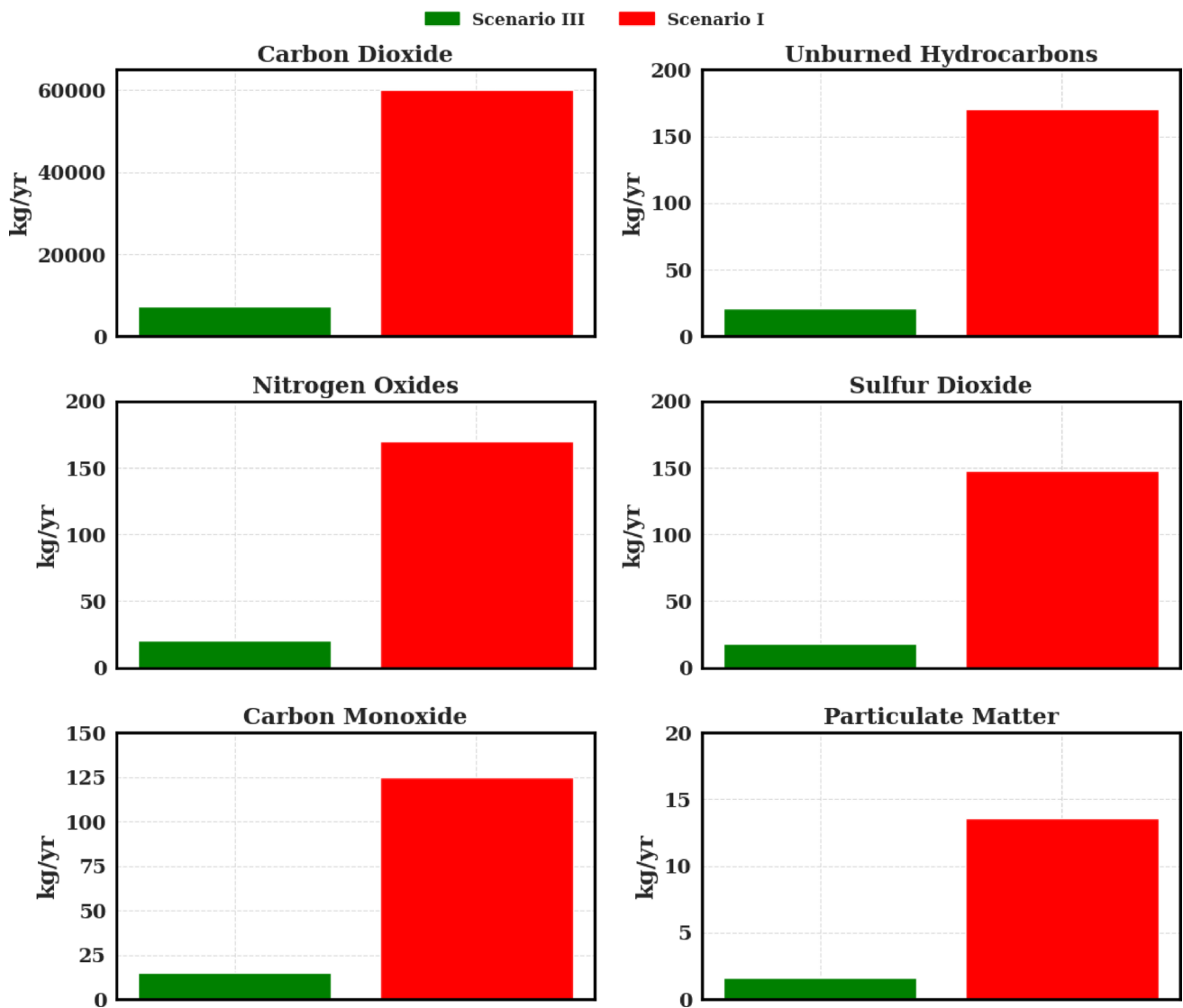


Fig. 10 Comparative analysis of the emissions for the two scenario including diesel generators

dioxide, other pollutants associated with diesel combustion were considered. The DG-only system emitted 125 kg/year of carbon monoxide (CO), 170 kg/year of unburned hydrocarbons (HC), 13.6 kg/year of particulate matter (PM), 148 kg/year of sulphur dioxide (SO₂), and 170 kg/year of nitrogen oxides (NO_x). These emissions were drastically reduced in the hybrid system, which produced only 15.1 kg/year of CO, 20.6 kg/year of HC, 1.64 kg/year of PM, 17.8 kg/year of SO₂, and 20.6 kg/year of NO_x, demonstrating a substantial improvement in local air quality and overall environmental performance. While the PV/Battery configuration (Scenario II) had no direct fuel-based emissions and thus appeared the cleanest operationally, its higher cost implication and inefficient energy production highlight trade-offs between environmental performance and system reliability. Although this study focused on operational

emissions, it is important to acknowledge that battery systems also pose environmental concerns over their lifecycle. Specifically, its end-of-life disposal or recycling strategy.

3.4 Sensitivity Analysis

The sensitivity analysis evaluated the impact of fluctuating diesel prices and solar irradiance on the performance of the HES, specifically Scenario III. The analysis considered a $\pm 25\%$ variation from the base values of these parameters to assess their influence on key economic and operational metrics. As illustrated in Fig. 11, increasing diesel prices from \$0.62/L to \$0.93/L led to a corresponding rise in the NPC and COE. For example, at an average solar irradiance level of 3.87 kWh/m²/day, the NPC increased from \$137,736 to \$144,435, while the COE rose from \$0.321/kWh to \$0.336/

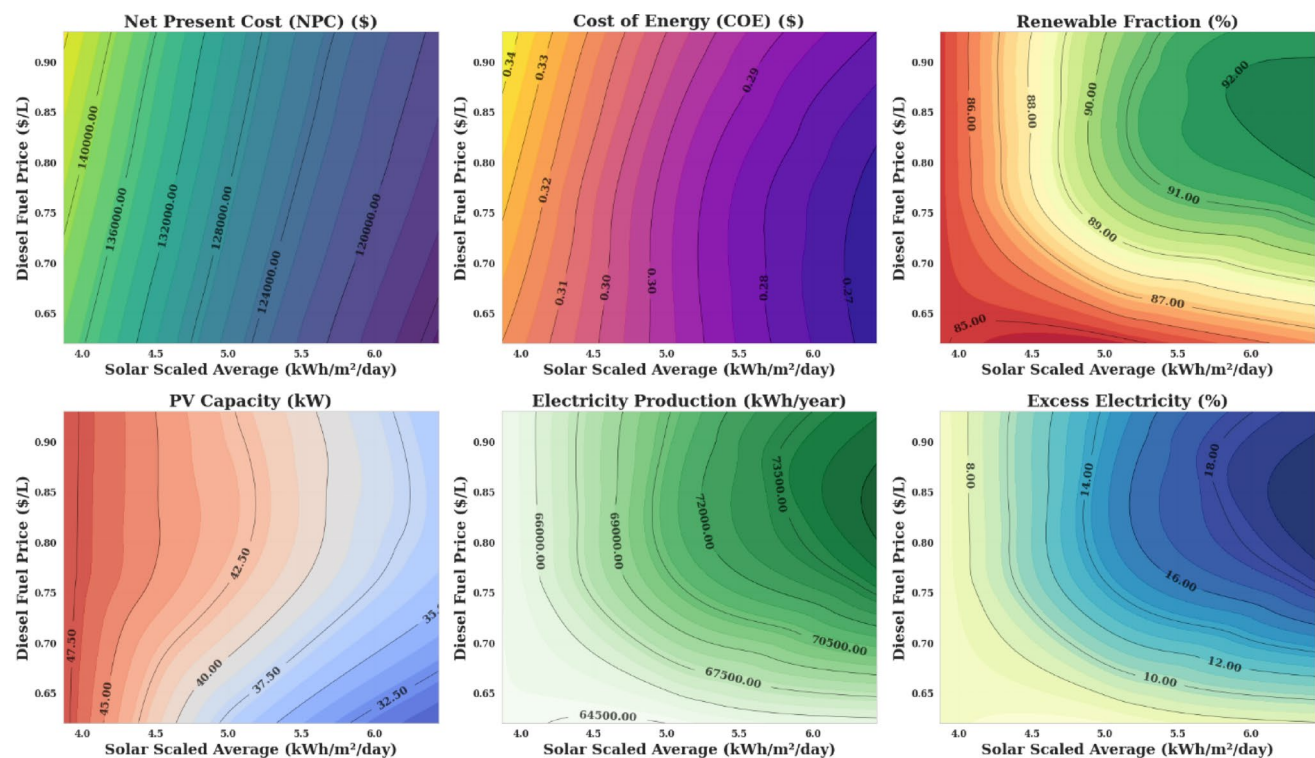


Fig. 11 Sensitivity of hybrid system performance to diesel price and solar irradiance variations

kWh. This reflects higher operational costs due to increased fuel expenses, which in turn encourage a shift towards greater reliance on solar energy and reduced DG usage. Diesel fuel consumption decreased from 2,740 L/year at \$0.62/L to 1,510.5 L/year at \$0.93/L under higher irradiance conditions (6.44 kWh/m²/day), demonstrating the system's adaptability in prioritising solar power to minimise cost.

Similarly, increases in solar irradiance significantly improved system performance. Electricity production rose from 64,757 kWh/year at 3.87 kWh/m²/day to 74,309 kWh/year at 6.44 kWh/m²/day, while the renewable fraction (RF) increased from 85.1% to 91.8%. These improvements reduced reliance on fossil fuels and enhanced sustainability. However, higher irradiance also led to an increase in excess electricity, from 6.63% to 18.79%, indicating a potential mismatch between generation and load demand. Besides, the study area is a semi-arid environment, where seasonal dust accumulation on solar panels can reduce actual irradiance reaching the panel surface, thereby affecting PV efficiency. While this factor was not explicitly modelled in the system design, the variability in irradiance captured in this sensitivity analysis provides insight into how environmental effects, such as dust can influence system performance.

In addition to resource sensitivity, further insight is demonstrated through a complementary financial sensitivity assessment. This analysis examined how variations in inflation and interest rates influence the performance of

the HES under Scenario III. As shown in the Fig. 12, system behaviour was assessed across a range of interest rates (18.75–29.81%) and inflation rates (15.90–32.77%). At low financial rates (18.75% discount, 15.90% inflation), the NPC and COE reduced to \$226,765.70 and \$0.22/kWh, respectively. Under higher rates (29.81% and 32.77%), the NPC increased to \$362,097.10, while the COE slightly declined to \$0.19/kWh. This reduction in COE reflects discounting effects rather than actual cost savings, as real expenditures remain elevated. To sustain system performance under these financial pressures, larger PV capacities, ranging from 48 to 54 kW, were required, which increased electricity production to 87,357 kWh/year and excess electricity to 31.02%. Despite these changes, the RF consistently exceeded 90%, affirming the system's resilience and environmental value under diverse economic conditions.

4 Implications of Hybrid Energy Deployment for Financial Inclusion

The results of this study demonstrate that HES can play a vital role in promoting financial inclusion, especially in remote and underserved regions like Giedam. Reliable access to electricity enables consistent operation of cash centres, which are crucial for providing banking services to rural populations. In the DG-only configuration (Scenario

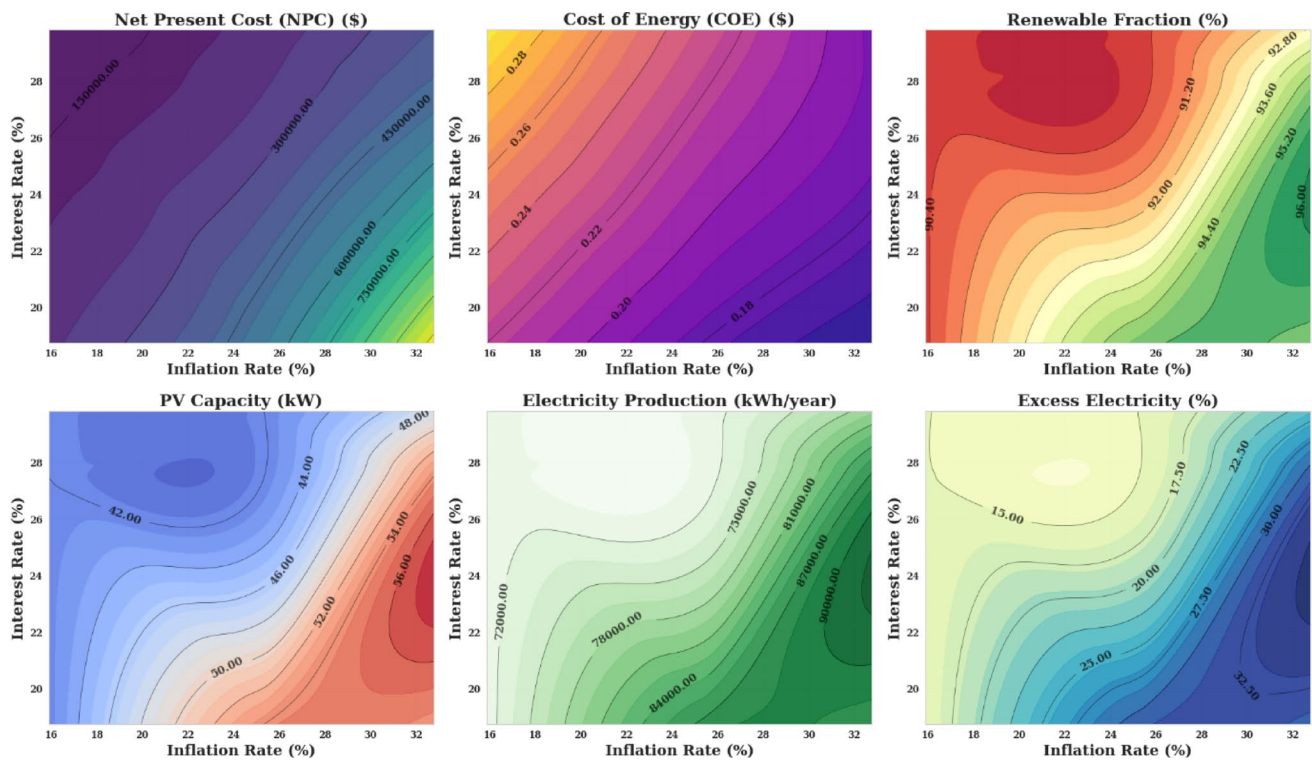


Fig. 12 Sensitivity of hybrid system performance to interest and inflation rate changes

I), although the energy demand was consistently met, inefficiencies such as 13.21% excess electricity and high diesel costs were evident. These operational challenges increase the running costs for FIs, which can lead to restricted service delivery, reduced operational hours, or even closure of cash centres in highly volatile fuel markets. Consequently, financial access becomes limited for the rural population. The PV/Battery system (Scenario II) offered cleaner energy but suffered from reliability challenges due to storage limitations, and variability in solar resources. Such unreliable scenario risks frequent service outages, undermining community confidence in digital financial systems. In contrast, the hybrid PV/DG/Battery configuration (Scenario III) achieved a balanced outcome, producing 64,756.80 kWh/year with no unmet load, minimal excess electricity (6.63%), and significantly reduced diesel dependence. By ensuring reliable, cost-effective power supply, hybrid systems support the uninterrupted functioning of cash centres, enabling more people to access savings, credit, digital payments, and other critical financial services. This reliability also encourages FIs to expand services in rural areas, thereby advancing financial inclusion.

While this study may not be all encompassing, the results suggest important policy considerations for various stakeholders. As a primary driver of the economy, governments and other development agencies should incentivise the deployment of hybrid renewable energy systems through

tax credits, grants, or concessional financing, particularly targeting FIs operating in off-grid areas. Energy infrastructure development should be formally integrated into financial inclusion strategies, recognising that reliable power is foundational to sustaining banking operations. Establishing standardised guidelines for renewable systems powering financial service points will also help ensure consistent performance and customer confidence. Besides, promoting public-private partnerships can accelerate investment and innovation in decentralised energy solutions for the financial sector. Such collaborations could pull private sector investment into funding mini-grids projects and RE-powered mobile banking hubs and microfinance centres in underserved regions. With all these measures, the role of renewable hybrid systems as enablers of financial access, economic growth, and social development in underserved communities can be strengthened.

5 Conclusions

This study demonstrates the potential of a hybrid solar photovoltaic (PV)-diesel energy system as a sustainable and cost-effective solution for improving energy reliability in remote Nigerian towns, with a representative focus on Giedam, a cash-dependent commercial hub. The analysis revealed that the hybrid PV-battery-diesel (Scenario III)

configuration outperformed diesel-only (Scenario I) and PV-battery systems (Scenario II) in terms of operational efficiency, environmental impact, and economic viability. The hybrid system achieved an 85% RF, reduced diesel fuel consumption by 88%, and minimised excess electricity to 6.63%. It effectively combined the reliability of diesel generators with the environmental and cost advantages of renewable energy, offering a balanced solution. Solar PV contributed 87.4% of the total energy production, while batteries provided 10.93 h of autonomy, enhancing the system's resilience to solar variability. Economic analyses confirmed the hybrid system's competitiveness, with the lowest COE and NPC among all scenarios. Sensitivity analyses highlighted the significance of fuel price fluctuations and solar resource variability, emphasising the need for adaptive strategies.

Based on these findings, we offer targeted recommendations for key stakeholders. Policymakers should introduce strategic subsidies and fiscal incentives, including tax breaks and import duty waivers, to support HES deployment in rural financial hubs. Additionally, they should integrate energy access with financial inclusion goals within national development policies, creating a unified approach to addressing both challenges. FIs should incorporate energy infrastructure planning into their rural banking strategies and form partnerships with renewable energy providers to co-finance hybrid systems in branch offices and cash centres. This would enhance service reliability and operational efficiency while supporting their sustainability goals. Energy planners and rural electrification agencies should prioritise hybrid systems in locations with high solar potential and inadequate grid coverage, designing modular and scalable solutions tailored to the specific load profiles of financial service operations.

The findings provide a practical framework for implementing HESs in rural cash centres, particularly in regions with abundant solar resources and unreliable grid power. By reducing reliance on diesel fuel, these systems lower greenhouse gas emissions while ensuring reliable energy access for essential cash-based services, fostering socio-economic development and environmental sustainability. Future research should explore the integration of advanced technologies, such as innovative energy storage solutions and demand-side management strategies, to further optimise system performance. Additionally, studies on financing mechanisms specifically designed for rural energy infrastructure would support broader implementation. This forward-looking approach would enhance the adaptability and long-term viability of HESs, reinforcing their critical role in supporting financial inclusion and achieving sustainable development goals in energy-deprived rural areas.

Author Contributions Joseph Yakubu Oricha: conceptualization, methodology, supervision, writing—original draft preparation. Sher-eefdeen Oladapo Sanni: conceptualization, methodology, simulation, writing—original draft preparation, Writing—reviewing and editing. Sikiru Mohammed Abdullahi: Writing—reviewing and editing.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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